Mohammad Abolnejadian

Applied AI Scientist, Verily Health; Master of Mathematics in Computer Science, University of Waterloo | ON, Canada

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Professional Experience

Applied AI Scientist Intern — May 2026 – Present

Verily Health (ex. Google Life Sciences) — *Waterloo, ON*

- Developing an agentic clinical framework grounded in clinical knowledge and self-improving agents, **reducing the development time for new agents** from months to weeks while increasing their auditability.
- Designed and prototyped a guideline-composing multi-agent pipeline that helps clinicians interactively develop clinical protocols grounded in trusted sources, **facilitating internal collaboration** between DS and clinical teams.
- Extended a clinical **knowledge graph** by curating and ingesting **1M+ clinical concepts**, their associations, and related encyclopedic knowledge, enabling a MedQA agent to achieve **90% accuracy**.

Graduate Research Engineer (Human-Centered AI) — Jan. 2025 – Present *(Expected Apr. 2027)*

University of Waterloo, CS HCI Lab — *Waterloo, ON*

- Led the R&D of an AI system that supports human decision-making during synchronous conversations by providing context-aware proactive glanceable insights, resulting in two first-author publications at CUI '25 and UIST '26.
- Developed a **multi-agentic RAG** workflow using **LangGraph** and Pydantic AI (structured output) with specialized decomposition and routing AI agents, integrating **Azure Foundry** for ASR and MCPs for visualization generation.
- Engineered a retrieval workflow by collecting **42,000+** data points via crawling and **scraping** constructing an **ETL pipeline** to embed unstructured data using GraphRAG mechanism for QFS retrieval.
- Prototyped PoCs by designing wireframes in **Figma** and implementing the system using **FastAPI** and **React** and deployed on **Google Cloud Run** to support mixed-methods user studies.

Research Assistant — Jun. 2023 – Oct. 2023

Tehran Institute for Advanced Studies (TEIAS) — *Tehran, Iran*

- Developed a personalized CS1 platform using **LLMs and in-context learning**; the AI-assisted group scored **28.3% higher on the third quiz** than traditional instruction, resulting in a first-author CHI '24 publication.
- Implemented and deployed the platform using **Flask**, **Flutter**, and **Firebase**; evaluated it in a comparative study with **85 high school students**. [Code]

Software Engineer — Jul. 2022 – May 2023

Cafe Bazaar — *Tehran, Iran*

- Collaborated in a **cross-functional agile** team to maintain a **10+ year legacy** in-app payment platform serving **50M+** users, reducing technical debt and migrating to new API schemas for consistency with internal microservices.
- Developed and deployed a **Django** backend with **RESTful APIs** and gRPC with Protobuf for a new gift card storefront using a stack including Docker and PostgreSQL to drive the platform's user engagement and retention goals.

Selected Technical Projects

SquatSage: Proactive Pose-Guided AI Strength Training Coach | *code*

- An on-device coaching application integrating **OpenCV**, a neural network classifier trained on **25K+** labeled data and deployed via **TensorFlow Lite**, and multimodal LLM to prevent injuries through real-time personalized feedback.

DistilKaggle: Large-Scale Computational Notebook Analytics Corpus | *code*, *dataset*

- An open-source dataset of **542K** Jupyter notebooks, condensed from **300GB+** to 3GB and extracted **34** code quality metrics through **large-scale data processing**, enabling accessible scientific usage; published at the MSR'24.
- Built an **ETL pipeline** and a **Random Forest** classifier for Kaggle performance tiers, achieving **57% F1**.

Technical Skills

Languages: Python, Java, JS, C/C++, SQL, Bash || **AI Agents/RAG:** LangChain, LangGraph, Pydantic AI, Prompt Engineering, ChromaDB, FAISS || **ML/Data:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy ||

Education

University of Waterloo, M.Math. (Thesis, Co-op) in Computer Science — Jan. 2025 – Present *(Apr. 2027)*

GPA: 4.0 | Notable Courses*: Multi-Agent Systems, Human-AI Interaction* — *Waterloo, ON*

Sharif University of Technology, B.Sc. in Computer Engineering — Sep. 2019 – Jul. 2024

GPA: 4.0 | Notable Courses*: Artificial Intelligence, Object-Oriented Design* — *Tehran, Iran*

Publications

2026 **InsightToast: Proactive Information Retrieval & Glanceable Visualization in the Side Channel of Data-Rich Meetings** **Mohammad Abolnejadian**, Matthew Brehmer. *ACM Symposium on User Interface Software and Technology (UIST '26), forthcoming.*

2025 **AInsight: Augmenting Expert Decision-Making with On-the-Fly Insights Grounded in Historical Data** **Mohammad Abolnejadian**, Shakiba Amirshahi, Matthew Brehmer, Anamaria Crisan. *ACM Conference on Conversational User Interfaces (CUI '25).*

2025 **MoTiCPS: Energy Optimization on Multi-Objective Task Scheduling in IoT-Integrated Cyber-Physical Systems** Abolfazl Younesi, Elyas Oustad, **Mohammad Abolnejadian**, Mohsen Ansari, Alireza Ejlali. *IEEE Transactions on Sustainable Computing, 10(4), 744–755.*

2024 **Leveraging ChatGPT for Adaptive Learning through Personalized Prompt-based Instruction: A CS1 Education Case Study** **Mohammad Abolnejadian**, Sharareh Alipour, Kamyar Taeb. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI '24).*

2024 **DistilKaggle: A Distilled Dataset of Kaggle Jupyter Notebooks** Mojtaba Mostafavi Ghahfarokhi, Arash Asgari, **Mohammad Abolnejadian**, Abbas Heydarnoori. *IEEE/ACM International Conference on Mining Software Repositories (MSR '24).*

2018 **Cyrus 2D Simulation Team Description Paper 2018** Nader Zare, Mohsen Sadeghipour, Ashkan Keshavarzi, Mahtab Sarvmeili, Amin Nikanjam, Reza Aghayari, Arad Firouzkoohi, **Mohammad Abolnejadian**, Sina Elahimanesh, Amin Akhgari. *RoboCup 2018, Soccer Simulation 2D League, Team Description Paper.*

Awards, Leadership, & Service

International RoboCup Competition Jun. 2018
2nd Place, Soccer 2D Simulation League, *Cyrus Team* *Montreal, QC*

Iran Open Robotics Competition Apr. 2018
1st Place, Soccer 2D Simulation League, *Cyrus Team* *Tehran, Iran*
1st Place, Soccer 2D Simulation League - Starter, *AE Team*

Graduate Student Representative Oct. 2025 – Present
School of Computer Science Council *Waterloo, ON*

- Voting member representing graduate students in the department's governance alongside faculty and school officers.

Teaching Assistant Sep. 2020 – Present
University of Waterloo | Sharif University of Technology *Waterloo | Tehran*

- Mentored students by conducting office hours and providing technical guidance on projects as Lead TA for Compiler Design course and TA for 10+ CS courses, including Software Design, Operating Systems, and Data Structures.
- Developed strong communication and leadership skills by designing and grading course projects and assignments as part of several teams of teaching assistants.

Mentor — Women in Mathematics Directed Research Program Jan. 2026 – Apr. 2026
University of Waterloo, Faculty of Mathematics *Waterloo, ON*

- Mentored four undergraduate students in the Women in Mathematics Directed Research Program through a guided research project, covering HCI fundamentals (mixed-methods analysis, user studies), data source identification and curation, and an introductory exposure to LLM and retrieval-augmented generation (RAG) concepts.